

much of miniaturisation or portability issues. In that case, the designer can afford to use bigger connectors since there are no major constraints in terms of compactness.

Significance of interoperability/compatibility. Another important aspect for designers is whether the connector supports interoperability and, if yes, with how many components or devices. If the given connector is compatible with products from three or four different vendors, the user can buy a connector made for the device for one brand and match it with the device of another brand. But a user needs to look into such prospects under worst-case scenarios.

When it comes to the manufacturing side, there are a lot of lead time and production issues. So the manufacturers have to look at the possibilities for replacements that can be used for a product in manufacturing.

Previously mentioned industry standards, such as USB and HDMI, help ensure intermateability of connectors.

Longevity. Longevity of a connector is generally defined by its insertion (and removal) cycles and gives an idea of how long it is expected to work properly. Insertion cycles are defined in the datasheet of each connector. Consider a connector with an insertion cycle of 200. Under normal conditions, you can insert and remove the connector approximately 200 times and still expect reliable performance. After that, the behaviour is not predictable.

For devices such as cellular phones, where we plug in the connector once or twice daily, the insertion cycle and longevity are expected to be greater.

Longevity depends on vibration aspects too; what wear and tear the product faces throughout its life cycle.

Environmental factors. The designer should keep in mind all the possible environmental conditions under which the device could be operated by the user, and accordingly ensure that the connectors and cables can withstand

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necessary requirements for EMI/EMC, RF, grounding, power and signal separation and many data-critical

these. For example, in case of fire or heat exposure, one should ensure that the connectors are of proper fire retardant (FR) grade. Sealed connectors are commonly used in such rugged environmental conditions.

Availability and serviceability. "Use of expensive or out of the ordinary connectors makes it difficult to service and satisfy customers," says T. Anand. But according to Gagan Shah, the product specialist of Phoenix Contact, "The selection criteria for connectors should not be based on the price. It is important to understand the need and application area of the customer before providing the right connector solution. Sometimes with additional cost the customer can get more benefits which cannot be achieved with low cost solutions."

The serviceable points of connectors should have proper detachment mechanism. Some connectors are plugable while others are directly soldered to the board. Using pin headers, wherever possible in the design, ensures ease of servicing.

Speed, capacity and other factors. Another criterion for selection of connectors is based on the kind of performance they can handle in terms of speed and response time. Choose connectors based on the standard ratings of voltage/current capacities and wire gauges they offer.

Determining the peak capacity is important as, if the device reaches peak consumption or has a current burst, it may no longer work reliably. The connector manufacturers typically define the specifications very clearly in their datasheet. If the tolerances are within the range of the particular component, the user will not have performance issues.

Grounding and shielding are

applications. Having snap-fit or locking sockets is another critical need for continuously or frequently moving devices. We now have double lock and triple lock facilities for connectors. The security standpoint can also be a factor—whether or not anybody can easily bypass into the communicated signal.

Standards to be fulfilled. Having appropriately certified cables and connectors (such as CE, UL and ISI) is a mandatory requirement for professional designs. Regulatory requirement such as Restriction of Hazardous Substances (RoHS) directive also needs to be fulfilled. In certain specific fields of application, like in the case of military uses, connectors used may require a specification such as MIL grade.

USB, HDMI, D-subminiature and/or DIN are the standards to be complied by I/O connectors for serial communication, modular jacks, audio jacks and RCA connectors.

Zero insertion force (ZIF) connectors can be used as on-board connectors for display modules or cameras, rectangular shrouded/unshrouded connectors, external cables and for testing and programming purposes.

Additionally, in some embedded systems for special applications, high-speed back-plane connectors, microprocessor and memory chips sockets and standard memory or SIM card connectors would also be required.

Though very rarely seen, sometimes there also exist a set of geospecific standards defined by countries or regulations. They insist on using a specific type of connectors for specific applications. ●

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